



13.3.4 Wrap-Up: Ticket Out the Door

1. The equation of circle W is $x^2 + (y + 4)^2 = 26$.

Select all the points that lie on circle W .

- ☐ (3,0)
- ☐ (5, -3)
- ☐ (1,1)
- ☐ (-3, -2)
- ☐ (-5, -5)

Unit 13, Lesson 4: Partitioning Line Segments



Benchmark

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.



Learning Target

- ☐ I can use coordinate geometry to partition a line segment.



13.4.1 Warm-Up: Heroic Person

1. Who do you consider heroic and inspirational?
2. Why did you pick this person?



13.4.2 Exploration: Partitioning a Line Segment

1. Directed line segment $\overline{SP}$ is shown.



- a. How many equal parts is $\overline{SP}$ partitioned into?

- b. Make a mark where $\overline{SP}$ is partitioned into a 2:3 ratio. Call this point R .

- c. Complete the statement.

Point R is

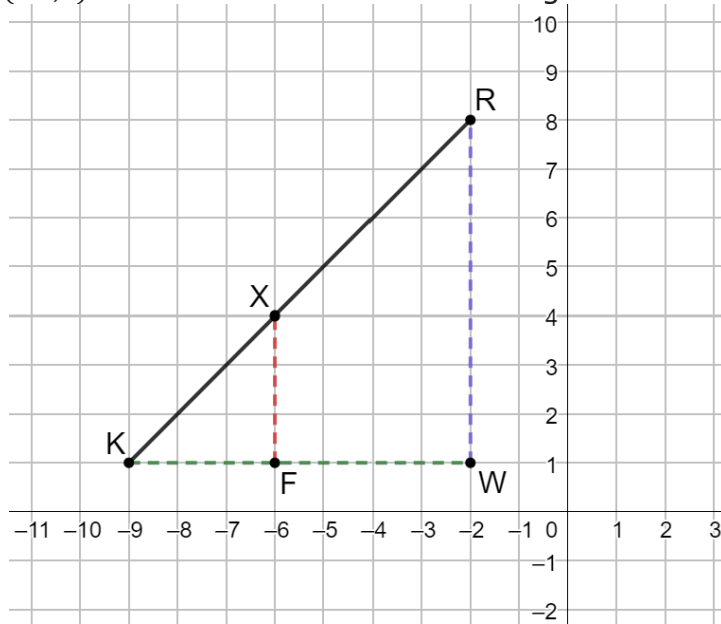
☐ $\frac{2}{3}$
☐ $\frac{2}{5}$
☐ $\frac{3}{5}$
☐ $\frac{3}{2}$

of the distance from S to P .

- d. Complete the equation that can be used to find the distance of $\overline{SR}$.

$$SR = \underline{\hspace{1cm}} SP$$

2. Point X partitions directed line segment KR in a 3:4 ratio. Points $K(-9,1)$, $F(-6,1)$, $X(-6,4)$, $R(-2,8)$, and $W(-2,1)$ are shown on the coordinate grid.



- a. Use the distances to write the numeric ratio for each segment ratio. Then, choose the type of ratio the segment ratio represents.

Segment Ratio	Numeric Ratio	Type of Ratio
$KF:FW$		☐ part:part ☐ part:whole
$KF:KW$		☐ part:part ☐ part:whole
$XF:RW$		☐ part:part ☐ part:whole
$KX:KR$		☐ part:part ☐ part:whole

- b. What do you notice about the numeric ratios?

- c. Two students were discussing the directed line segment KR .

Nevaeh	Isabella
Point R is a dilation of point X centered at point K using a scale factor of $\frac{3}{4}$.	Point R is a dilation of point X centered at point K using a scale factor of $\frac{3}{7}$.

Work with your partner to explain why neither student has the correct scale factor. Include in your explanation what the correct scale factor should be so that the result of dilating at point K is point R .

- d. Discuss with your partner which ratios can be used to justify that $\triangle KXF \sim \triangle KRW$. Summarize your discussion.
- e. With your partner, write a conjecture on how point X could be found if only given the endpoints of the directed line segment KR and the ratio.

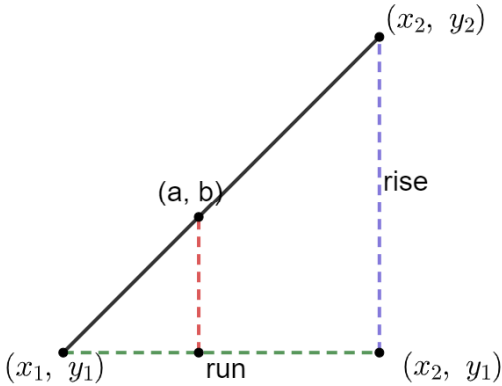
- f. Ask two other pairs of partners for the conjecture they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

Conjecture	Initials

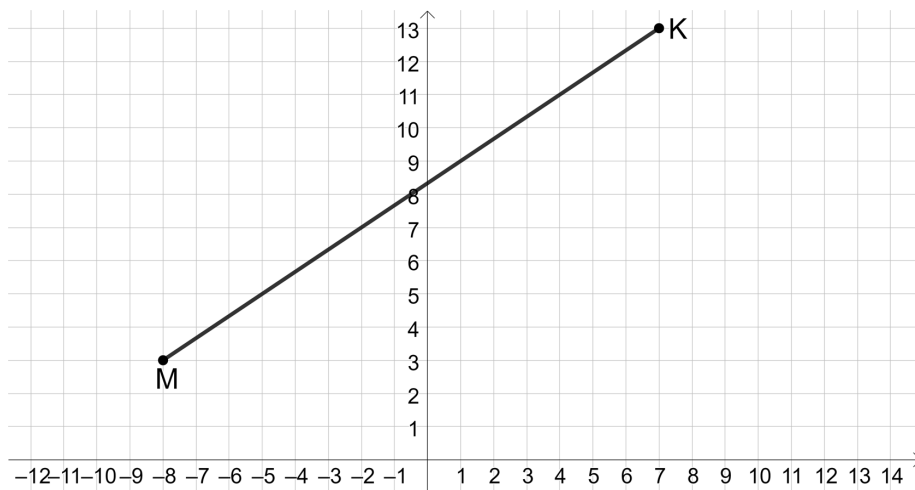


13.4.3 Guided Instruction: Partitioning a Line Segment

- Generalize the conjecture in the Exploration if the given ratio is part:whole.



2. Directed line segment MK is shown with endpoints at point $M(-8,3)$ and point $K(7,13)$.



Directed line segment MK is partitioned by point B in a ratio of 2:3.

- Write an equation that can be used to find the distance of $\overline{MB}$.
- Use your equation to find the horizontal distance of $\overline{MB}$.

- Use your equation to find the vertical distance of $\overline{MB}$.
- Use the horizontal distance and the vertical distance to plot point B on $\overline{MK}$.
- What are the coordinates of point B ?

3. Directed line segment XY has endpoints at point $X(-6,1)$ and point $Y(4,4)$.



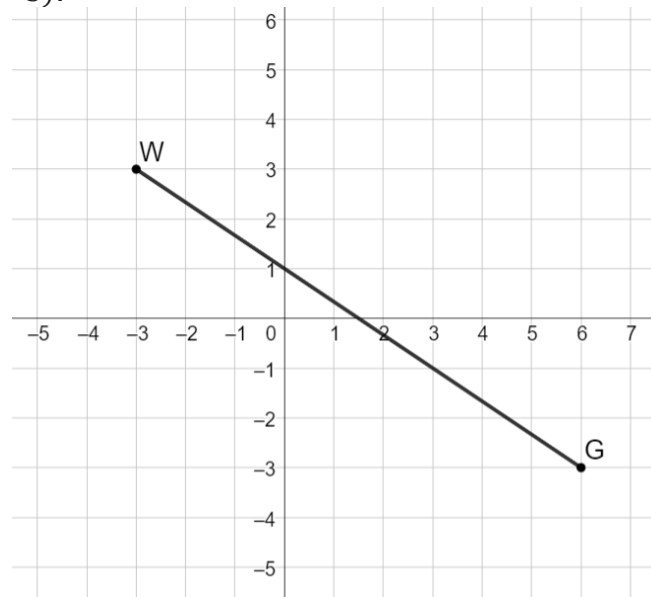
Directed line segment XY is partitioned $\frac{1}{2}$ of the distance from X to Y at point W .

- What is the horizontal distance of $\overline{XW}$?
- What is the vertical distance of $\overline{XW}$?
- What are the coordinates of point W ?



13.4.4 Try It: Partitioning a Line Segment

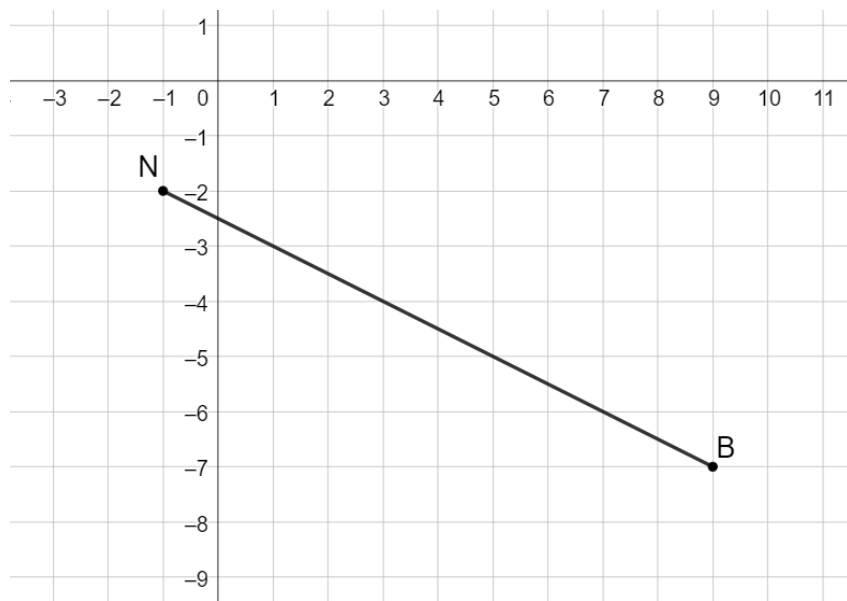
1. Directed line segment WG has endpoints at $W(-3,3)$ and $G(6,-3)$.



Directed line segment WG is partitioned $\frac{2}{3}$ of the distance from W to G at point D .

- What is the horizontal distance of $\overline{WD}$?
- What is the vertical distance of $\overline{WD}$?
- What are the coordinates of point D ?

2. Directed line segment BN has endpoints at point $B(9, -7)$ and point $N(-1, -2)$.



Directed line segment BN is partitioned by point H at a 4:1 ratio.

What are the coordinates of point H ?



13.4.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.